

Dynamical Systems - Lab Exam Notes

Find the general solution of a differential equation:

```
eq:=a*diff(x(t),t)+b*x(t)=0
dsolve(eq,x(t))
```

Find the solution of an IVP:

```
eq:=a*diff(x(t),t)+b*x(t)=0
dsolve({eq,x(0)=a},x(t))
```

```
for 2nd grade:
dsolve({eq,x(0)=a,D(x)(0)=b},x(t))
```

Find the general solution of a linear nonhomogeneous equation. Example:

$$x'(t) + x(t) = 2e^t - 7e^{-3t} \rightarrow x(t) = \frac{7e^{-3t}}{2} + e^t + e^t c_1$$

```
dsolve(eq,x(t)) --> x(t)=xp+xh where xh contains a constant c1
```

Find a diagonalizable matrix A (but not diagonal) with given eigenvalues:

1. find P invertible real matrix
2. introduce J diagonal matrix with the given eigenvalues
3. compute $A = P \cdot J \cdot P^{-1}$

Find general solution of differential system $X' = AX$ (where A is known or given):

```
X(t):=<x(t),y(t),x(t)>
dsolve(diff(X(t),t)=A.X(t),X(t))
```

Find the matrix of the linearized non-linear system around an equilibrium point (x_0, y_0) :

1. write the Jacobian matrix of the non-linear system: $J:=\text{Jacobian}([f1(x,y),f2(x,y)], [x,y])$
2. find the linearization matrix: $A:=\text{subs}([x=x0,y=y0],J)$
3. linearized x is $\text{diff}(x(t),t)=A.x(t)$ and linearized y is $\text{diff}(y(t),t)=A.y(t)$
4. linearized system is $\langle \text{diff}(x(t),t), \text{diff}(y(t),t) \rangle = A \cdot \langle x(t), y(t) \rangle$

Find the general solution of the differential equation of the orbits of a system:

ex: system $\dot{x} = y, \dot{y} = -4\sin x$ equation of the orbits: $\frac{dy}{dx} = -\frac{4\sin(x)}{y}$ (y is a function of x) in Maple:

```
eq_orbits:=diff(y(x),x)=-4*sin(x)/y(x)
dsolve(eq_orbits,y(x))
```

Check H is a first integral:

```
H:=(x,y)->a*x^2+b*y^2
eq:=diff(H(x,y),x)*f1+diff(H(x,y),y)*f2
check if eq == 0
```

Check $(0,0)$ is hyperbolic - check if $Re(\lambda_1) \neq 0$ and $Re(\lambda_2) \neq 0$ (where λ_1 and λ_2 are the eigenvalues of the Jacobian matrix at equilibrium point):

```
J:=Jacobian([f1(x,y),f2(x,y)], [x,y])
J_0,0:=eval(J,[x=0,y=0])
Eigenvalues(J_0,0)
```

Euler method formula with step size $h = \frac{x^*-x_0}{n}$:

$$y_{k+1} = y_k + h * f(x_k, y_k)$$
$$x_{k+1} = x_k + h$$

Improved Euler method formula:

$$y_{k+1} = y_k + \frac{h}{2}f(x_k, y_k) + \frac{h}{2}f(x_k + h, y_k + h * f(x_k, y_k))$$
$$x_{k+1} = x_k + h$$

If φ is the exact solution of the IVP, then y_k is the approximation of $\varphi(x_k)$.

primitive = integral

Bessel function of first kind = $BesselJ(v, x)$ = Bessel functions describe the radial part of vibrations of a circular membrane Bessel's equation: $x^2 y'' + xy' + (-v^2 + x^2)y = 0$

Error function = $erf = erf(x) = \frac{2(\int_0^x e^{-t^2} dt)}{\sqrt{\pi}}$

Solution curve = particular integral curve which is the solution

Check that column vector u is an eigenvector corresponding to the eigenvalue λ : check $A \cdot u = \lambda \cdot u$

Complex numbers: i in Maple is I Fixed points: $f(x)=x$ Equilibrium points: $f(x)=0$

Let linear system $\dot{X} = AX$ with $\det(A) \neq 0$. Then the system is a:

- NODE when λ_1, λ_2 real have the same sign
- SADDLE when λ_1, λ_2 real have opposite signs
- CENTER when $\lambda_{1,2} = \pm i\beta$ with $\beta > 0$
- FOCUS when $\lambda_{1,2} = \alpha \pm i\beta$ with $\alpha \neq 0$ and $\beta > 0$

If system is a center or saddle, it has a global first integral.

Let linear system $\dot{X} = AX$ and equilibrium point $(0,0)$.

- if $Re(\lambda_1) < 0$ and $Re(\lambda_2) < 0$ - $(0,0)$ attractor
- if $Re(\lambda_1) > 0$ and $Re(\lambda_2) > 0$ - $(0,0)$ repeller
- if system is a center - $(0,0)$ stable
- if system is a saddle - $(0,0)$ unstable